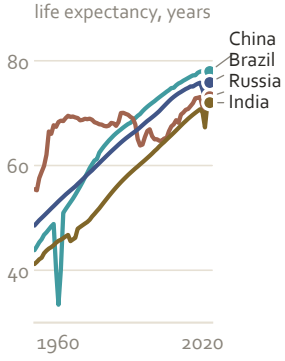
The palette survives colour-vision deficiency because lightness alternates

Six tones and a four-line chart, light twin · simulated after Machado, Oliveira and Fernandes (2009), severity 1

Normal vision



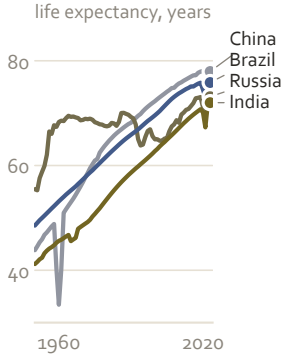


smallest neighbour distance

18.3 floor 15

Protanopia



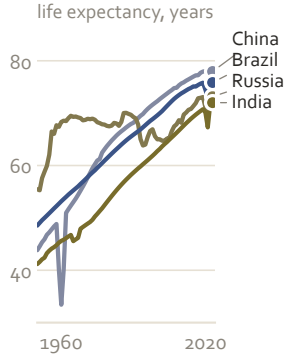


smallest neighbour distance

12.3 target 8

Deuteranopia



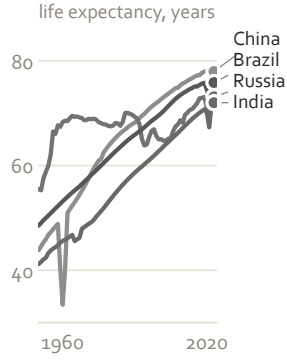


smallest neighbour distance

11.4 target 8

Greyscale





smallest neighbour distance

6.4 lightness alone

Data: UN WPP (2024) and others, processed by Our World in Data (CC BY 4.0); simulation: generators/colorlab.py · cvd_sheet.py · git dabag b8